Tunger Hong

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EDUCATION

Bachelor of Science — *Carnegie Mellon University*
Electrical and Computer Engineering — *expected May 2027*
GPA: 3.37/4.0

High School Diploma — *Wayzata High School*
2019-2023

GPA: weighted 4.6, ACT: 36

TECHNICAL SKILLS

Programming Languages	Python, C++/Vulkan, C, x86 Assembly, Java, Matlab, and System Verilog
Operating Systems	Mac OSX, Linux, and Windows

WORK EXPERIENCE

Boston Scientific — *May 2025 - Aug 2025*
Software Intern — *Saint Paul, MN*

- Engineered and optimized data-processing pipelines for heart-failure monitoring and chemical-sensor studies, improving analysis throughput and reliability.
- Redesigned the file-storage architecture and built a robust Python library to ingest, process, and visualize CSV outputs.
- Created calibration workflows and analytical visualizations, enabling the research team to rapidly interpret and act on experimental results.
- Developed the backend foundation for an interactive data-visualization web application, designed for scalability and seamless integration with future database systems.

Carnegie Mellon University — *Jan 2026 -*
Teaching assistant for 15-362 Computer Graphics — *Pittsburgh, PA*

- Delivered a lecture on Bidirectional Reflectance Distribution Functions (BRDFs) to a class of 50 students, covering physically-based rendering and light interaction models.
- Designed and delivered weekly recitations on core course topics such as path tracing and rasterization, and assisted in making recitation slides and material.

Carnegie Mellon University — *Jun 2024 - Dec 2024*
Research Assistant — *Pittsburgh, PA*

- Conduct research in auditory psychology, with a specialization in the ventriloquist effect, examining the interplay between auditory and visual stimuli in perception.
- Leverage advanced data analysis techniques and tools, including NumPy and pandas, to extract valuable insights from experimental data.

COURSEWORK

15-472 Real Time Computer Graphics — *Spring 2026*
Created a state-of-the-art renderer with Vulkan and C++. Implemented real-time 3D watercolor stylization, animation handling, various material types, lights, and shadow mapping.

15-362: Computer Graphics — *Fall 2025*
Implemented rasterization, mesh manipulation, pathtracing, and animation within Scotty3D, a 3D modeling, rendering, and animation package.

18-213 Computer Systems — *Fall 2024*
Learned x86 assembly and implemented low-level system software such as malloc, shell, and a web proxy using C.